Computational aspects of penalized least-squares trend estimation with controlled smoothness

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Abstract

Penalized least-squares trend estimators provide an interpretable way to separate persistent variation from short-term fluctuations. Their practical use nevertheless requires choices about difference order, reference differences, smoothing intensity, and numerical implementation. We examine these choices through a reproducible computational framework for finite-difference penalties. We derive the polynomial null space and the spectral representation of the smoother, and normalize the trace-based smoothness index so that its range is independent of the difference order. We also study the conditional profile estimator of a nonzero reference-difference mean, distinguishing this derivation from procedures available in the software. Controlled experiments compare naive and cached dense linear systems, a reusable singular-value decomposition, and a banded solver. We separately investigate repeated full-trend reconstruction and scalar effective-degrees-of-freedom evaluations, including numerical agreement and preprocessing time. A nonlinear synthetic example illustrates the behavior of classical smoothing criteria, while monthly atmospheric carbon-dioxide observations demonstrate controlled smoothness at different penalty orders. The results highlight workload-dependent computational trade-offs rather than the universal superiority of any individual algorithm. The accompanying Python implementation and executable experiments provide a transparent basis for further numerical and statistical investigation.

Keywords: Whittaker–Henderson smoothing, spectral decomposition, smoothing parameter selection, time-series trend, reproducible computation

1 Introduction

An observed time series may combine low-frequency behavior with temporary movements and measurement noise. Estimating its trend is a descriptive task: the resulting curve depends on how fidelity to the observations is traded against smoothness. Whittaker's graduation framework and subsequent finite-difference penalized least-squares (PLS) estimators formalize this compromise Whittaker (1923); Guerrero (2007). The Hodrick–Prescott filter is an important second-difference special case, but the statistical problem admits other penalty orders and nonzero reference values for the differences Cortés-Toto et al. (2017).

The tuning parameter must also be chosen. Cortés-Toto, Guerrero, and Reyes Cortés-Toto et al. (2017) investigate attained smoothness when choosing the penalty with cross-validation (CV), generalized cross-validation (GCV), corrected Akaike information criterion (AICc), or Bayesian information criterion (BIC). Their simulations vary trend shape, series length, variability, and seasonality. Efficient Whittaker–Henderson computation has an extensive history, including methods exploiting banded factorization and structure to reduce numerical expense Weinert (2007); consequently, diagonalization is not presented here as a newly discovered mathematical algorithm.

In the scientific-computing literature, Balam et al. Balam et al. (2025) compare normal equations, QR, and singular-value decomposition (SVD) in least-squares computation and discuss conditioning. Azofeifa and Moreles Azofeifa and Moreles (2025) present a different numerical model, but similarly distinguish the formulation, solver, numerical comparisons, and limitations. We similarly distinguish statistical formulation, numerical algorithms, empirical comparisons, and their limitations.

Our objective is to provide a small, auditable Python package, `trendestimation`, that implements the *published* PLS family and exposes examples as executable notebooks within Sphinx documentation. Specifically, we describe the affine solution for general order d and reference mean μ, its spectral implementation, the criteria available in the public API, and numerical experiments that report computational time, solver agreement, and precomputation costs alongside reproducible illustrations. Experimental forecast-horizon cross-validation and dynamic tracking of minima from other research projects are not included. The distinction between an in-sample smoother and an ex ante forecast is maintained throughout.

2 Model and spectral representation

2.1 Finite-difference penalized least squares

Let $y = (y_1, \ldots, y_n)^\top \in \mathbb{R}^n$ denote an equally spaced series. For an integer $1 \le d < n$, let $D_d \in \mathbb{R}^{(n-d)\times n}$ represent the dth forward-difference operator; for example, row entries of D_2 are $(1, -2, 1)$. We consider the published estimator

$$\widehat{\tau}_{\lambda,\mu} = \arg\min_{\tau\in\mathbb{R}^n} \left\{ \|y - \tau\|_2^2 + \lambda \|D_d\tau - \mu\mathbf{1}\|_2^2 \right\}, \qquad \lambda \ge 0, \tag{1}$$

where $\mu \in \mathbb{R}$ is a fixed reference mean for the differences. The choice $\mu = 0$ gives the familiar pure penalty. The implementation additionally permits $d = 0$, with $D_0 = I_n$, as a well-defined edge case; the geometric statements below concern $1 \le d < n$.

Write $Q_d = D_d^\top D_d$ and choose $p \in \mathbb{R}^n$ satisfying $D_d p = \mu \mathbf{1}$. Using zero-based indices $t = 0, \dots, n-1$, one convenient choice is $p_t = \mu \binom{t}{d}$ because $D_d \binom{t}{d} = 1$. Setting $x = \tau - p$ and differentiating the strictly convex objective gives

$$\widehat{\tau}_{\lambda,\mu} = p + H_\lambda (y - p), \qquad H_\lambda = (I_n + \lambda Q_d)^{-1}. \tag{2}$$

Indeed, $(I_n + \lambda Q_d)x = y - p$ and $I_n + \lambda Q_d$ is positive definite for every finite $\lambda \ge 0$. This affine expression is equivalent to the normal equations $(I_n + \lambda Q_d)\widehat{\tau} = y + \lambda \mu D_d^\top \mathbf{1}$.

2.2 Proof of the *d*-dimensional kernel

For $1 \le d < n$, the dth forward difference has the explicit binomial stencil

$$(D_d \tau)_t = \sum_{k=0}^{d} (-1)^{d-k} \binom{d}{k} \tau_{t+k}, \quad t = 0, \dots, n-d-1. \tag{3}$$

The first $n-d$ columns of D_d form an upper triangular $(n-d) \times (n-d)$ submatrix with nonzero diagonal $(-1)^d$. Hence $\operatorname{rank}(D_d) = n - d$. As $Q_d = D_d^\top D_d$, $x^\top Q_d x = \|D_d x\|^2$, and therefore

$$\ker Q_d = \ker D_d, \qquad \dim \ker Q_d = n - \operatorname{rank}(D_d) = d. \tag{4}$$

The vectors $1, t, \dots, t^{d-1}$ have vanishing dth forward differences and are linearly independent on $n > d$ nodes. Thus they give a basis of the kernel, proving that exactly d eigenvalues of Q_d vanish in exact arithmetic. Under $\mu = 0$, the limit $\lambda \to \infty$ is the least-squares projection onto polynomials of degree $\le d - 1$. For instance $d = 1$ preserves a constant, $d = 2$ a line, and $d = 3$ a quadratic. When $\mu \ne 0$, the limiting set is instead the affine space $\{\tau : D_d \tau = \mu \mathbf{1}\}$, which has the same d homogeneous directions but includes a degree-d particular solution. In the implementation $p_t = \mu \binom{t}{d}$.

2.3 Interpretation of higher orders and the spline connection

For a sufficiently smooth function f observed on an equally spaced mesh of width h, Taylor expansion yields $\Delta_h^d f(x_t) = h^d f^{(d)}(x_t) + O(h^{d+1})$. Consequently, under appropriate mesh-refinement assumptions,

$$h^{1-2d} \sum_t (\Delta_h^d f(x_t))^2 \longrightarrow \int |f^{(d)}(x)|^2 \, dx. \tag{5}$$

This is a mathematical bridge to roughness penalties in smoothing splines. A Whittaker-type trend smoother penalizes differences of fitted ordinates directly, while a P-spline instead penalizes differences of coefficients in a chosen spline basis; these

are related ideas but not identical estimators. For a continuous penalty weight λ_{cont}, the corresponding discrete coefficient is scaled approximately as $\lambda_{\text{disc}} = \lambda_{\text{cont}} h^{1-2d}$.

The order d is an integer representing the order of the derivative being approximated; it is not a continuous interpolation between derivative orders. Comparing different d values is a change of regularization model, so equal numerical λ does not imply equal smoothing. It is useful to examine orders $d = 1, \ldots, 5$ under a common normalized trace-based index; however, arbitrary orders and polynomial nullspaces have long been known Weinert (2007); Biessy (2026). Larger d can be ill-conditioned because the binomial stencil has growing coefficients and the positive eigenvalues may span many orders of magnitude. Neither higher d nor higher polynomial degree guarantees a better trend or numerical stability.

2.4 Null space, SVD, and limiting trends

The rows of D_d are independent in exact arithmetic, so $\operatorname{rank}(D_d) = n - d$ and $\dim \ker D_d = d$. The null space consists of the sampled polynomials of degree at most $d-1$. Let $D_d = U \Sigma V^\top$ be a full SVD. With nonnegative eigenvalues δ_j of Q_d, ordered so that exactly d are zero, the smoother becomes

$$H_\lambda = V \operatorname{diag}\left(\frac{1}{1+\lambda\delta_1}, \ldots, \frac{1}{1+\lambda\delta_n} \right) V^\top. \tag{6}$$

Therefore the transformed observations $c = V^\top (y - p)$ can be computed once, and every new penalty value changes only the scalar attenuation factors $(1 + \lambda\delta_j)^{-1}$. At $\lambda = \infty$ the exact limiting fit is $p + P_{\ker D_d}(y - p)$, rather than a computation using an arbitrarily large finite penalty.

For fixed μ, the effective degrees of freedom are $\operatorname{EDF}(\lambda) = \operatorname{tr}(H_\lambda) = \sum_j (1 + \lambda\delta_j)^{-1}$. We report Guerrero's trace-based smoothness $S_G = 1 - \operatorname{tr}(H_\lambda)/n$ alongside the normalized coordinate

$$S(\lambda) = \frac{n - \operatorname{tr}(H_\lambda)}{n - d} \in [0, 1], \quad S(0) = 0, \quad S(\infty) = 1. \tag{7}$$

For $1 \le d < n$, this coordinate is strictly increasing for finite λ because $S'(\lambda) = (n-d)^{-1} \sum_{\delta_j > 0} \delta_j / (1 + \lambda\delta_j)^2 > 0$. The normalization changes the scale of the original index but not the ordering of smoothness levels Guerrero (2008).

3 Classical parameter selection and numerical implementation

3.1 Selection criteria

For a fixed numerical value of μ, define $e_\lambda = y - \widehat{\tau}_{\lambda,\mu}$, $\operatorname{RSS}_\lambda = \|e_\lambda\|_2^2$, $h_{ii} = (H_\lambda)_{ii}$, and $\nu_\lambda = \operatorname{tr}(H_\lambda)$. The published classical selectors studied in Cortés-Toto et al. (2017)

are represented, up to constants irrelevant to an optimizer, by

$$\mathrm{CV}(\lambda) = \frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_{\lambda,i}}{1-h_{ii}}\right)^2, \tag{8}$$

$$\mathrm{GCV}(\lambda) = \frac{\mathrm{RSS}_\lambda/n}{(1-\nu_\lambda/n)^2}, \tag{9}$$

$$\mathrm{AICc}(\lambda) = \log(\mathrm{RSS}_\lambda/n) + \frac{2\nu_\lambda+1}{n-\nu_\lambda-2}, \tag{10}$$

$$\mathrm{BIC}(\lambda) = \log(\mathrm{RSS}_\lambda/n) + \frac{\nu_\lambda \log n}{n}. \tag{11}$$

The software checks denominators and excludes undefined candidates, evaluates a declared positive grid of penalty values, and compares an exact infinite-penalty candidate. In particular, a near-interpolating BIC minimum can be driven by the log(RSS) boundary, as our example illustrates. We do not interpret a finite-grid BIC minimum close to zero as a universally reliable statistical choice.

Guerrero Guerrero (2007) also considers the feasible plug-in $\widehat{\mu} = \mathrm{mean}(D_d y)$. This option is available *for fitting*; however, the formula for leave-one-out CV in (8) presumes that μ remains fixed under deletion. The current public selectors intentionally require a fixed numerical μ. Likewise, $\mathrm{tr}(H_\lambda)$ is a conditional EDF and is not automatically the full degrees of freedom of a procedure that estimates μ from the same observations.

3.2 Smoothing correction, inversion, and exact endpoints

With the full orthogonal eigenbasis of the penalty, $Q_d = V\,\mathrm{diag}(\delta_1,\ldots,\delta_n)V^\top$, the $n-d$ positive eigenvalues are $\delta_j = \sigma_j(D_d)^2$, and the remaining d are zero. Consequently, the spectral eigenvalues of $H_\lambda = (I+\lambda Q_d)^{-1}$ are

$$\alpha_j(\lambda) = \begin{cases} (1+\lambda\delta_j)^{-1}, & 1 \le j \le n-d, \\ 1, & n-d < j \le n. \end{cases} \tag{12}$$

This exposes both the exact limiting polynomial directions and the scalar attenuation of the penalized directions. If $\nu(\lambda) = \mathrm{tr}(H_\lambda) = d + \sum_{j=1}^{n-d}\alpha_j(\lambda)$, the original trace-based Guerrero index

$$S_G(\lambda) = 1 - \frac{\nu(\lambda)}{n} \quad \text{ranges from } 0 \text{ to } 1-\frac{d}{n}. \tag{13}$$

The upper bound is $1-d/n$, *not* $1-n/d$. The library also uses a normalized correction:

$$S(\lambda) = \frac{S_G(\lambda)}{1-d/n} = \frac{n-\nu(\lambda)}{n-d} \in [0,1]. \tag{14}$$

The map is strictly increasing for $\lambda \in [0, \infty)$ because

$$S'(\lambda) = \frac{1}{n-d} \sum_{j=1}^{n-d} \frac{\delta_j}{(1+\lambda\delta_j)^2} > 0. \tag{15}$$

The exact endpoints are $S(0) = 0$ and $S(\infty) = 1$; for any interior target, a one-dimensional bracketed solver inverts this map. Equal normalized smoothness makes trace scales comparable across d; it does not make the underlying trend geometries or roughness penalties identical.

3.3 Where the computational acceleration comes from

For fixed μ put $p_t = \mu\binom{t}{d}$ and $c = V^\top(y - p)$. One common projection computes c; after that, for each candidate λ,

$$\operatorname{tr}(H_\lambda) = d + \sum_{j=1}^{n-d} \alpha_j(\lambda), \tag{16}$$

$$\|y - \widehat{\tau}_{\lambda,\mu}\|^2 = \sum_{j=1}^{n-d} (1 - \alpha_j(\lambda))^2 c_j^2. \tag{17}$$

Thus a sweep of M values can obtain trace, normalized smoothness, GCV, AICc, BIC and related scalar profiles in $O(Mn)$ *after* the shared eigendecomposition and $O(n^2)$ projection. A full reconstructed trend remains $O(n^2)$ per penalty, and exact ordinary CV leverage diagnostics also involve the spectral basis. The full dense SVD has $O(n^3)$ preprocessing time and $O(n^2)$ storage. This decomposition/reuse trade-off explains the potential responsiveness of repeated-criteria explorations without claiming unconditional superiority to banded linear solvers Weinert (2007); Biessy (2026).

Large batches of spectral attenuation operations are naturally vectorizable. Dense linear algebra can also benefit from optimized multithreaded BLAS on CPUs. In principle, a GPU backend could parallelize matrix decompositions and batched matrix operations, but host/device transfer, synchronization, and initialization costs may negate a benefit for short series. We found `numpy` and `scipy.linalg.svd` in the package currently reviewed, not a released GPU solver; therefore no GPU speedup, power saving, or measured device execution is reported. A future benchmark should record CPU and GPU models, precision, BLAS and accelerator versions, transfer time, warmups, memory, and the number of candidate penalties.

3.4 Reusable computation and cost

The package exposes a reusable `TrendSmoother(n, order=d)` object. Construction forms the difference operator and its SVD. For each candidate λ, a fitted curve is obtained with the transformed vector c and spectral attenuation in (6). The library offers `fit`, `degrees_of_freedom`, `smoothness`, and `select_lambda`, while plotting

utilities visualize fits and criterion profiles. Sphinx incorporates executable Jupyter notebooks, with deterministic random seeds and figure code for inspection.

The computational trade-off must distinguish decomposition from reuse. A dense full SVD costs $O(n^3)$ preprocessing and $O(n^2)$ memory; after projection, EDF, smoothness, GCV, AICc, and BIC summaries cost $O(n)$ per penalty candidate, while full reconstruction and exact CV leverage scores require $O(n^2)$ each. In contrast, the banded structure of $I_n + \lambda Q_d$ allows efficient specialized direct solvers Weinert (2007). Thus the phrase *fast computation* means repeated spectral evaluation under a stated workload, not a blanket claim that SVD beats banded factorization. A separate reproducible benchmark in the repository measures setup and fit times independently.

4 Gaussian marginal likelihood and profiling the difference mean

4.1 Likelihood selection for fixed μ

A probabilistic interpretation of the penalty assumes observation noise $\varepsilon \sim N(0, \sigma^2 I_n)$ and stochastic differences $D_d\tau = \mu\mathbf{1}+\eta$ with $\eta \sim N(0, \sigma^2\lambda^{-1}I_{n-d})$. The d unpenalized polynomial coordinates act as nuisance fixed effects. In the $r = n - d$ positive-penalty singular coordinates $z_j = v_j^\top y$, write $p_t^{(1)} = \binom{t}{d}$, $b_j = v_j^\top p^{(1)}$. Marginalization gives

$$z_j \mid \mu, \lambda, \sigma^2 \sim N\left(b_j\mu, \sigma^2 c_j(\lambda)\right), \quad c_j(\lambda) = 1 + \frac{1}{\lambda\delta_j}. \tag{18}$$

This is a Gaussian residual/marginal-likelihood interpretation, not a new form of PLS. It is related to likelihood-based Whittaker–Henderson smoothing Biessy (2026). Profiling the variance σ^2 for a fixed numerical μ leaves, up to an additive constant,

$$J(\lambda; \mu) = r \log\left(\frac{1}{r}\sum_{j=1}^{r}\frac{(z_j - b_j\mu)^2}{c_j(\lambda)}\right) + \sum_{j=1}^{r}\log c_j(\lambda). \tag{19}$$

An independently developed branch of the package, `feature/mle-smoothness-and-data-mu-20261021`, adds this profiled Gaussian score for fixed μ or the conditional plug-in $\mu = \overline{D_d y}$. That plug-in must not be mislabeled as a joint MLE.

4.2 A closed-form estimator of μ conditional on λ

Let $w_j(\lambda) = c_j(\lambda)^{-1} = \lambda\delta_j/(1 + \lambda\delta_j)$. For a fixed positive λ, the generalized least-squares optimum satisfies

$$\widehat{\mu}(\lambda) = \frac{\sum_{j=1}^{r} w_j b_j z_j}{\sum_{j=1}^{r} w_j b_j^2}. \tag{20}$$

The denominator is positive because $D_d p^{(1)} = \mathbf{1}$. Equation (20) is also obtained by minimizing the PLS objective jointly in τ and μ for a fixed λ. Substitution in (19)

Table 1 Selected penalty and normalized smoothness for the case in Fig. 1. Trend RMSE is computed against the known latent simulation only.

Criterion	$\widehat{\lambda}$	S	Trend RMSE
CV	6.17	0.777	0.242
GCV	6.14	0.777	0.242
AICC	9.74	0.805	0.248
BIC	2.45e-07	0.000	0.454

gives the one-dimensional concentrated score $J(\lambda; \widehat{\mu}(\lambda))$, whose optimization produces a joint profile-likelihood candidate $(\widehat{\lambda}, \widehat{\mu})$. Such a solution depends on the Gaussian variance model, may lie near a boundary, and must be checked for identifiability and other local minima. This paper provides the equation and a separately reproducible numerical test; joint μ profiling is not yet an exposed release API.

In fact, taking the limit $\lambda \downarrow 0$ using $w_j \sim \lambda\delta_j$ proves

$$\lim_{\lambda\downarrow 0} \widehat{\mu}(\lambda) = \frac{(p^{(1)})^\top Q_d y}{(p^{(1)})^\top Q_d p^{(1)}} = \frac{\mathbf{1}^\top D_d y}{n-d} = \overline{D_d y}. \tag{21}$$

For positive λ the two expressions differ in general. Estimating μ adds a rank-one adjustment to the data-to-trend linear operator, so the fixed-μ effective degrees of freedom and ordinary leave-one-out denominators are not automatically valid after joint estimation. Prior literature already considers nonzero difference means Guerrero (2007); Cortés-Toto et al. (2017); the implementation contribution here is an explicit, testable profiling workflow, not a claim to have first introduced μ or likelihood selection.

5 Reproducible numerical illustration

5.1 Source-inspired nonlinear design

To demonstrate the interface without introducing an external data source, we reproduce the nonlinear simulation family of Cortés-Toto et al. Cortés-Toto et al. (2017):

$$\tau(t) = 0.6 f_{\text{Beta}(30,17)}(t/n) + 0.4 f_{\text{Beta}(3,11)}(t/n), \quad t = 1, \ldots, n. \tag{22}$$

We take $n = 50$, add independent normal errors with standard deviation 0.5, omit seasonality, and fix the pseudo-random seed at 19. These choices define one illustrative replication, not a full reproduction of the article's factorial simulation. We fit $d = 2$, $\mu = 0$, and select λ from the same 121-point geometric grid augmented with the exact $\lambda = \infty$ candidate under each criterion. The latent trend in (22) is known, allowing an *ex post* trend-recovery RMSE; it is not a future forecast error.

As shown in Table 1, CV and GCV agree in this particular realization, whereas AICc selects a somewhat smoother fit. BIC selects a penalty extremely close to zero on the inspected finite grid; its apparent interpolation tendency illustrates a known

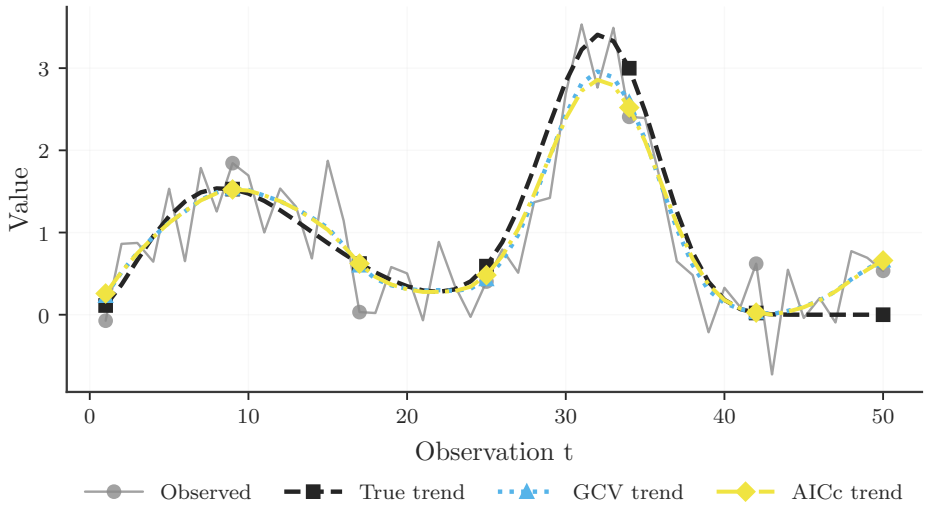


Fig. 1 A synthetic nonlinear case based on Cortés-Toto et al. Cortés-Toto et al. (2017). The curves show *in-sample* estimates and known simulated truth, not out-of-sample forecasts. Source data are regenerated by `figures/make_figures.py`.

practical boundary concern and is not evidence that BIC is generally ineffective. The simulated recoveries are descriptive; no general method ranking follows from a single seed.

5.2 Algebraic and numerical checks

An independent check compares the SVD implementation with the direct normal equations for $n = 50$, $\lambda = 30$, $d \in \{1, 2, 3\}$ and $\mu \in \{0, 0.15\}$ on a second fixed-seed Gaussian sample. The maximum componentwise absolute difference is 2.01×10^{-12} across these six cases, and the limiting constraint $D_d \widehat{\tau}_{\infty,\mu} = \mu \mathbf{1}$ holds to maximum error 4.42×10^{-12}. Both values result from the supplied deterministic regression script in IEEE double precision; they verify the tested algebra rather than establish stability for all n, d, λ. Higher differencing orders can be ill-conditioned, and systematic machine-dependent speed measurements remain a distinct stage of the study.

6 Measured computational cost and repeated parameter evaluation

6.1 Experimental design and independent baselines

Numerical correctness and elapsed time are distinct properties. We therefore benchmark the same PLS estimator at matched (n, d, μ) and identical candidate penalties, using four implementations: (i) a deliberately naive dense solver that reconstructs D_d,

Table 2 Leading arithmetic complexity for M reconstructed trends of length n and fixed order d. Setup is counted and no exact operation counts or GPU measurements are asserted.

Method	Setup	Repeated full fits
Fresh dense solves	rebuilt per λ	$O(Mn^3)$
Cached dense Q_d	dense penalty matrix	$O(Mn^3)$
Spectral SVD	$O(n^3)$	$O(Mn^2)$
Direct banded solver	$O(nd^2)$	$O(Mnd^2)$

Table 3 Host-measured end-to-end median times in milliseconds, including one-time setup, for M full fitted trends. Every method computes the same trends for the same n, $d = 2$ and $\mu = 0$; R is the naive-fresh-dense time divided by the spectral time. Values and the numerical parity check are generated from the raw CSV files, not entered manually.

n	M	Fresh	Cached	Spectral	Banded	R
40	1	0.029	0.027	0.204	0.123	0.14
40	64	1.388	0.739	1.302	1.032	1.07
100	1	0.075	0.073	0.801	0.281	0.09
100	64	4.589	2.642	2.013	1.532	2.28
200	1	0.471	0.350	3.580	0.535	0.13
200	64	20.984	10.348	4.982	2.017	4.21
400	1	3.192	3.217	20.165	1.103	0.16
400	64	196.554	106.894	20.885	3.188	9.41

$Q_d = D_d^\top D_d$, and solves the normal equations at every candidate; (ii) a denser but fairer baseline that caches Q_d and repeats its dense solve; (iii) the reusable full-SVD spectral implementation of `TrendSmoother`; and (iv) a banded solver that assembles Q_d directly in its sparse-band storage and uses a structure-aware linear solver. The last comparison is crucial: beating dense normal equations does not imply beating specialized PLS solvers Weinert (2007).

For every case a deterministic Gaussian sequence is generated, μ is held at either zero or a fixed nonzero value, and λ spans 10^{-3} to 10^3 on a geometric grid. The `quick` experiment varies $n \in \{40, 100, 180\}$, $M \in \{1, 16, 64\}$ with $d = 2$ and adds $d \in \{1, 3, 5\}$ at $n = 100$, $M = 24$, together with a nonzero-μ case. The `full` experiment reaches $n = 400$, $M = 256$ and additional difference orders. All results include one-time setup (not only post-SVD solves); we record medians, extrema, and median absolute deviations over repeated independent runs. We bound maximum disagreement against an independent banded solve for *every* tested penalty. Runs use a declared random seed, source revision, machine metadata, NumPy/SciPy versions and a stated BLAS-thread limit. The code and machine-readable outputs are confined to `paper/benchmarks/`.

Table 4 Median end-to-end times (milliseconds) to evaluate M effective-degrees-of-freedom traces, including setup. These entries measure only scalar $\mathrm{tr}(H_\lambda)$ quantities, not reconstructed trends.

n	M	Fresh	Cached	Spectral	Banded
40	8	0.213	0.150	0.203	0.322
100	8	1.059	0.832	0.801	1.072
100	32	4.220	3.307	0.885	3.615
180	32	15.914	12.208	2.830	10.114
250	64	92.614	59.421	6.090	38.391

6.2 End-to-end full-fit timing results

The crossover between methods should be evaluated using *end-to-end* times, including the SVD setup cost, for both a single smoothing parameter and a repeated-λ sweep. A ratio greater than one denotes a speed advantage over the naive baseline on the reported machine, not a universal speedup. In particular, the structured banded baseline can outperform a reusable SVD for full trend reconstruction.

6.3 Scalar smoothness and EDF are different workloads

To evaluate $\mathrm{tr}(H_\lambda)$ repeatedly, the spectral method needs only $n - d$ positive eigenvalues:

$$\mathrm{tr}(H_\lambda) = d + \sum_{j=1}^{n-d} (1 + \lambda\delta_j)^{-1}.$$

This is $O(Mn)$ work *after* spectral setup, not $O(Mn^2)$ reconstructed trends. An independent second experiment compares repeated dense computations of $\mathrm{tr}[(I + \lambda Q_d)^{-1}]$, a banded solver applied to all n canonical right-hand sides, and the spectral sum. The experiment checks algebraic equivalence separately from timed regions. Even when the spectral trace is the most efficient *marginal* score after setup, the $O(n^3)$ SVD cost can still dominate small workloads.

Measurement provenance: Platform: Windows-10-10.0.26100-SP0; CPU: Intel64 Family 6 Model 183 Stepping 1, GenuineIntel; Logical CPUs: 32; NumPy: 2.4.6; SciPy: 1.17.1; BLAS threads: 1; Repetitions: 7; Preset: full; Revision: local-unrecorded.
Maximum observed absolute trend disagreement: $4.32e-10$; maximum EDF trace disagreement: $5.48e-12$.

6.4 Interpreting speed and computational cost

A timed speed ratio is defined by the two end-to-end medians, for example $R_{\mathrm{dense/spectral}} = t_{\mathrm{dense}}/t_{\mathrm{spectral}}$. A value above one favors the spectral approach for the tested workload, not for arbitrary data lengths or hardware. CPU time, wall time, arithmetic complexity and energy consumption are different quantities. CPU numerical libraries already use optimized native linear algebra, but this experiment does

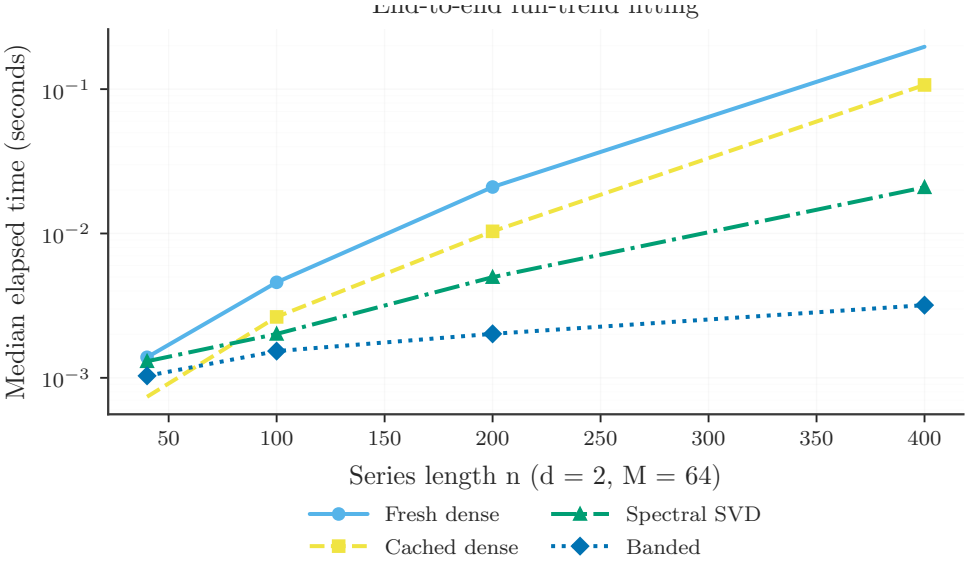


Fig. 2 End-to-end median times for $M = 64$ fitted curves, including setup, generated on the executing host from the raw CSV files. These are hardware-specific measurements, not an assumption that spectral reconstruction is the universally fastest method.

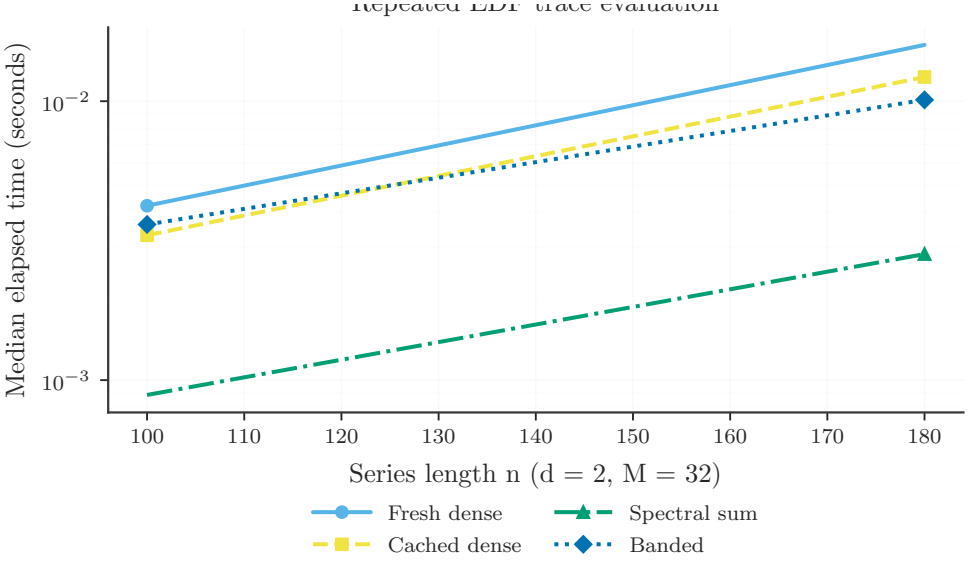


Fig. 3 Repeated effective-degrees-of-freedom trace evaluations ($M = 32$), including preprocessing. The quantity is scalar $\mathrm{tr}(H_\lambda)$, not a full trend reconstruction.

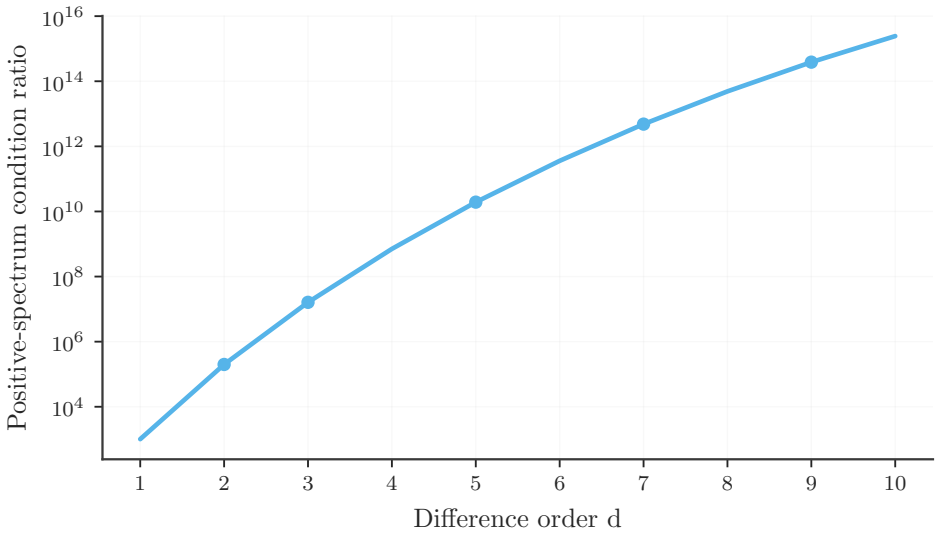


Fig. 4 Condition ratio of the positive spectrum of $D_d^\top D_d$ for $n = 50$, $d = 1, \ldots, 10$. Values are recomputed from singular values of D_d; they do not represent hardware-specific runtime.

not establish GPU execution or electricity use. An explicit device and power measurement would be necessary to make either claim. Results are intentionally generated only through a manual command, so the manuscript does not inadvertently contain timings from a different machine presented as those of its authors.

7 A controlled exploration of larger difference orders

The software accepts any integer $0 \leq d < n$, but this algebraic admissibility must be distinguished from floating-point stability. We therefore conduct a deterministic diagnostic with $n = 50$, seed 24021, difference orders $d = 1, \ldots, 10$, $\mu = 0$, and common normalized smoothness $S = 0.86$. At each order, we form the full SVD of D_d, compute the condition ratio

$$\kappa_+(Q_d) = \frac{\max_{1 \leq j \leq n-d} \delta_j}{\min_{1 \leq j \leq n-d} \delta_j}, \tag{23}$$

and compare the spectral fitted trend with an independent direct solution of $(I + \lambda Q_d)\tau = y$. The ratio is calculated from the *strictly positive* penalty eigenvalues; the full Q_d has zero eigenvalues and is singular for every $d \geq 1$.

The ratio rises from about 10^3 at $d = 1$ to 2.44×10^{15} at $d = 10$. At $d = 10$, the maximum componentwise difference between the SVD-based fit and the direct normal-equation solution is approximately 3.36×10^{-7} for the declared sample and smoothing target, versus approximately 1.14×10^{-15} at $d = 1$. The direct normal-equation result is not presumed an exact ground truth at large condition numbers: both approaches are affected by finite precision. The experiment illustrates why support

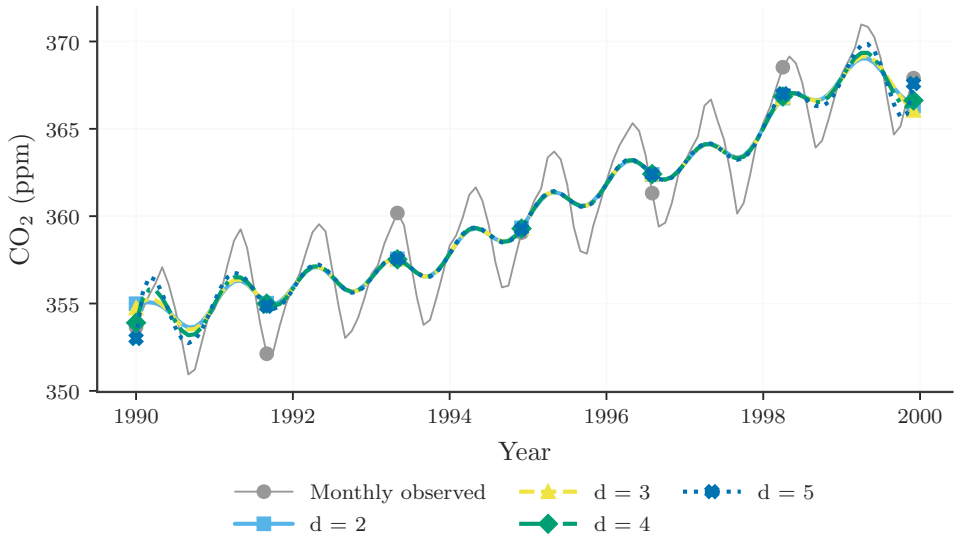


Fig. 5 Historical Mauna Loa atmospheric carbon dioxide, monthly aggregates of weekly measurements from 1990–1999, and two-sided PLS fits with common normalized smoothness $S = 0.86$. These are descriptive trends, not forecasts.

for many difference orders needs accompanying conditioning diagnostics, rather than establishing new high-order theory, unconditional stability, or performance superiority. Machine-specific setup/fitting times are separately exported by the experiment script for later audit; they are not promoted to general speed benchmarks.

8 Observed-data illustration: atmospheric carbon dioxide

A practical use for trend estimation is the Mauna Loa atmospheric CO_2 record. These are physical observations, not a synthetic series. We use the historical weekly concentrations distributed by `statsmodels.datasets.co2` Keeling and Whorf (2004); Seabold and Perktold (2010), aggregate weekly data to calendar-month means for 1990–1999, and obtain $n = 120$ equally spaced monthly values. No monthly values are missing within this window. The original monthly sequence mixes an increasing long-run component with strong seasonal variation; the latter must not be interpreted solely as instrument noise.

We estimate descriptive low-frequency curves for $d = 2, 3, 4, 5$. Because a conventional GCV grid can select a nearly interpolating solution in this seasonally varying case, we instead prescribe the clearly defined normalized trace-based index $S = 0.86$ and solve Eq. (14) for λ at each order. The resulting penalties are, respectively, 37.03, 134.01, 488.32, and 1702.30. Even at the same normalized S, distinct d values preserve different polynomial directions and may have different endpoint behavior.

Table 5 Direct joint normal-equation versus spectral profile-μ numerical check; independent Gaussian test series, $n = 50$, $\lambda = 27$. The values are maximum absolute floating-point differences, not inferential errors.

d	$\lvert\Delta\mu\rvert$	$\lVert\Delta\tau\rVert_\infty$
1	1.02×10^{-16}	1.17×10^{-15}
2	1.49×10^{-16}	7.61×10^{-15}
3	2.65×10^{-14}	8.04×10^{-14}
5	2.21×10^{-12}	2.64×10^{-11}

For $d = 2$ at the corresponding $\lambda = 37.03$, the observed-difference plug-in is $\overline{D_2 y} = 0.002119$ ppm/month^2, while the likelihood-profile formula gives $\widehat{\mu}(\lambda) = -0.003092$ in the same units. This illustrates that a positive-penalty profile estimate is not generally the raw observed-difference mean. Profiling λ jointly over the declared positive search interval selected a very small value (approximately 6.14×10^{-6}) in this particular dataset; seasonal structure is an important modeling limitation and such boundary behavior should not be interpreted as evidence of superior estimation. The figure therefore uses the separate controlled-smoothness choice.

A full atmospheric analysis would explicitly model annual seasonality, measurement uncertainty, and revisions to the underlying data. The present package smooths the complete observed series; it does not estimate predictive skill from future held-out years.

9 Conclusions

We have presented an auditable implementation of established penalized least-squares smoothing, together with additional derivations of the difference nullspace, the corrected smoothness index, and a Gaussian mean-difference profile estimator. The latter has numerical checks but is not yet a released joint-MLE API. Our Mauna Loa observations give a real, explicitly descriptive application of high-order differencing. Spectral decomposition makes repeated penalty evaluation transparent and inexpensive after preprocessing; specialized banded solvers remain important competitors for isolated fits and large samples. The source-inspired experiment shows how the same observations can lead to distinct smoothness choices under established criteria, while deterministic checks support correctness in a bounded set of cases. The separate numerical comparison places naive dense normal equations, cached dense solves, reusable SVD, and an independently assembled banded solver under matched workloads. This distinction is important: spectral reuse is particularly attractive for repeated trace-based scores, but is not automatically optimal for reconstructing full trends. Broader performance conclusions require reported machine characteristics, workload-specific runtimes, and stability diagnostics. Neither high-order differencing

nor likelihood-based smoothing is claimed as a newly discovered concept. No GPU acceleration has been verified in the released CPU implementation.

Declarations

Funding. To be confirmed by the authors before submission.
Competing interests. To be confirmed by the authors before submission.
Ethics approval and consent. Not applicable to the presented numerical algorithms and historical aggregate time-series examples; authors to verify.
Data availability. The simulated data can be reproduced from the supplied scripts and seeds. The observed monthly carbon-dioxide series is derived from `statsmodels.datasets.co2`; data access and provenance are described in the manuscript.
Code availability. The implementation and paper-specific simulation and benchmark scripts are maintained at https://github.com/heri-espino/trendestimation. A permanently archived release identifier should be added before submission.
Author contributions. To be confirmed and approved by both authors before submission.

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